

Putting windows in the black box

Developing a set of diagnostic and analytical tools to support multilateral price indexes in production

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Checkout the project site (examples, paper, code, and more!)

1 Motivation

- Multilateral methods take as input a large number of records and involve a lot of calculation. Notably:
  - GEKS family - multilayered bilaterals
  - TPD and TDH - large scale regression (typically with sparse matrices)
- Hard to analyze** - Unlike traditional (bilateral) methods, not intuitive to understand and explain the drivers of price change
- Opaqueness introduces risk** - lack of understanding also implies it is hard to detect issues

NSOs must validate that everything operates effectively & be able to undersand (and explain) movements.

**Goal of the project:** Define and demonstrate representative analytical outputs and supporting diagnostics with open data to (1) lead to a consensus and (2) simplify adoption

2 What to look out for with multilateral methods?

- Potential issues to look out for depend on the class of multilateral:
- TDH - estimated using product characteristics
  - TPD/GEKS - uses appropriate product id (i.e. homogeneous product)

Characteristics or products thus need to be consistently defined for the length of the window.

Main risk: if coding of (1) characteristics or (2) product ids change within the window -> an incorrect movement will occur.

3 Defining diagnostics

3.1 Monitoring/validation

**Goal:** Check that the data was received and is reasonable

Examples:

- Revenue by store / retailer
- Number of unique products (or row)
- Average unit value per unique product code
- Quantities and prices aligned to historic trends

3.2 Input diagnostics

**Goal:** Check that the data is structured appropriately for input to multilateral indices, and errors have been followed up.

Specifically:

- Churn in unique products (could be due to data issues, misclassification, or other coding/processing issues).
- Missingness in product characteristics
- Unit prices and quantities lie within historic averages

3.3 Analytical/output diagnostics

**Goal:** Understand the quality adjusted price change & pick up on issues (whether with the data or the estimation process).

Example analytical outputs:

- Decompose into 'point effects'
- Combine with other characteristics (brand, type of produce, etc)

Example output diagnostics:

- Validating products that contribute over an expected threshold.
- Compare movement for previous periods between this run and previous runs

3.4 When is it enough

It is important to note that an 80-20 rule is probably feasible. Not all possible scenarios can be forecasted, but common dashboards should answer most questions (or alert for further investigation).

Structured (and well documented) data pipeline key to support analysts to investigate further as needed.

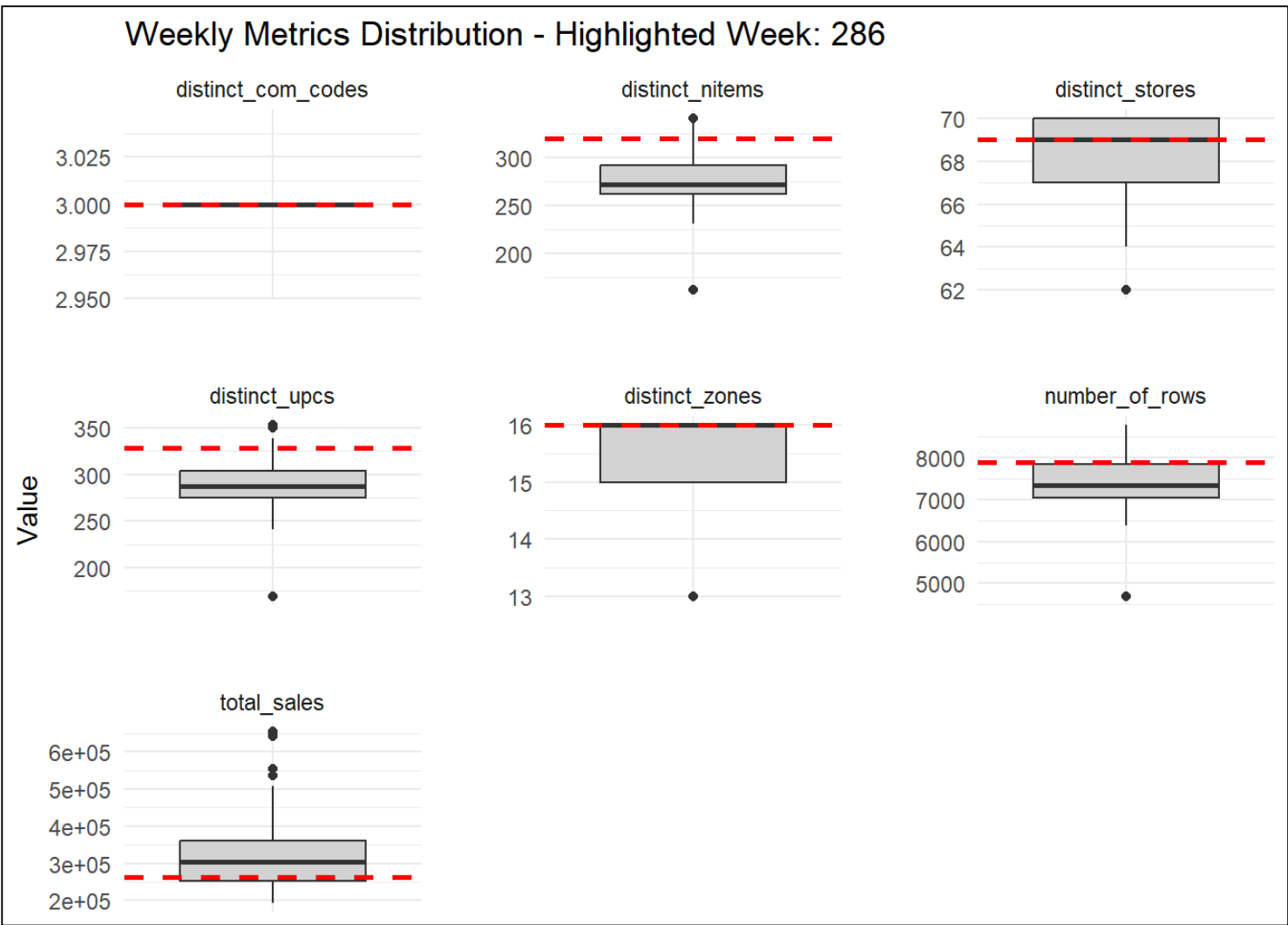
5 Example with Dominick's

5.1 Monitoring/validation

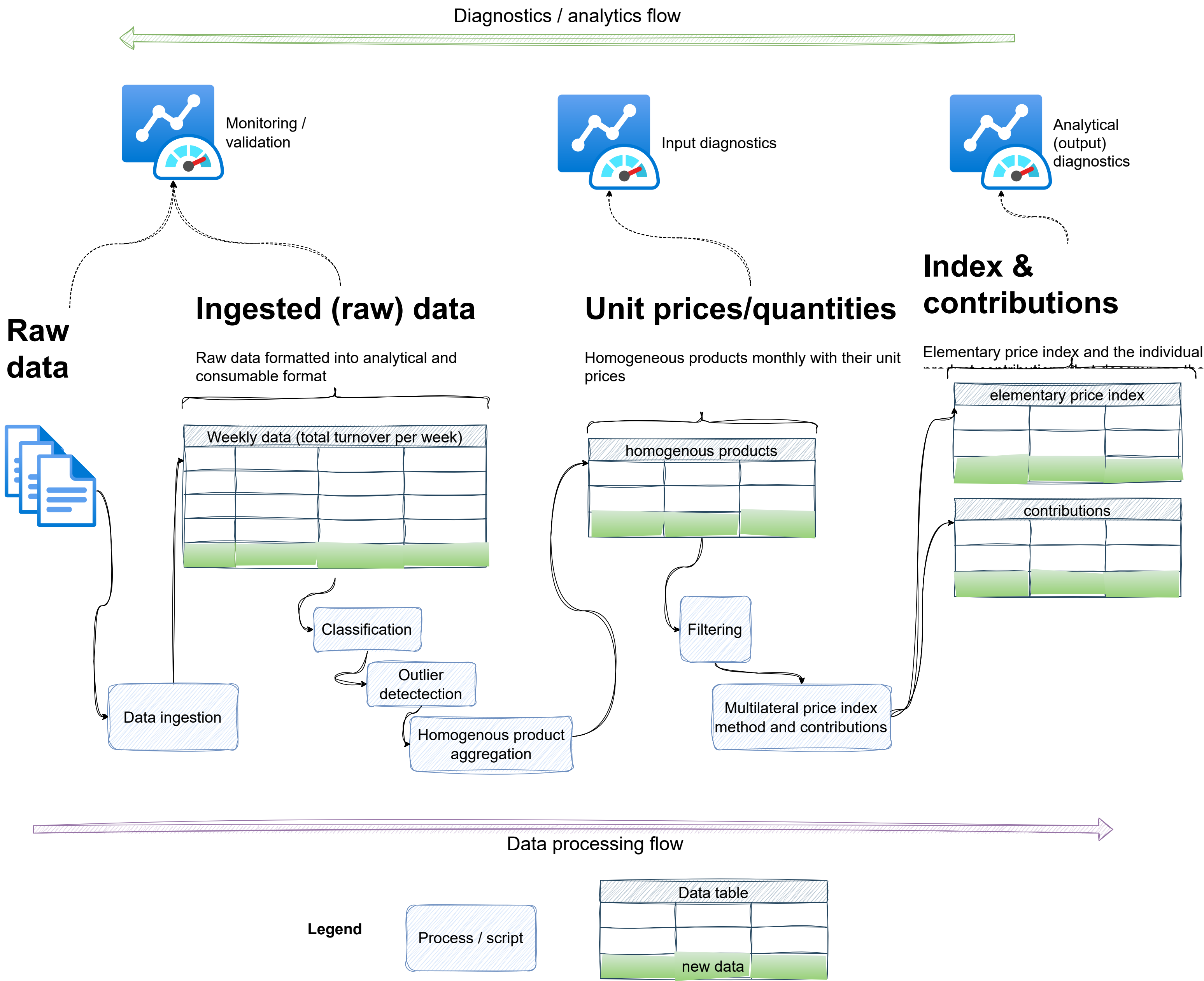
Example weekly summary

| Column visibility ▾ |      |                |             |            |            |
|---------------------|------|----------------|-------------|------------|------------|
|                     | WEEK | number_of_rows | total_sales | week_start | week_end   |
| 1                   | 286  | 7880           | 263621.3    | 1995-03-02 | 1995-03-08 |
| 2                   | 287  | 8181           | 310166.82   | 1995-03-09 | 1995-03-15 |
| 3                   | 288  | 8098           | 296978.96   | 1995-03-16 | 1995-03-22 |
| 4                   | 289  | 8144           | 275520.61   | 1995-03-23 | 1995-03-29 |

Example statistics per week



4 How to operationalize



Example data flow:

- Step 1: Raw (weekly) data files are received.
- Step 2: Raw data files are loaded into internal data store that supports simpler analytics.

Both will power monitoring and validation diagnostics.

- Step 3: Classification, outlier detection and aggregation by homogenous products (i.e. aggregation across time, geography, and products) is carried out.

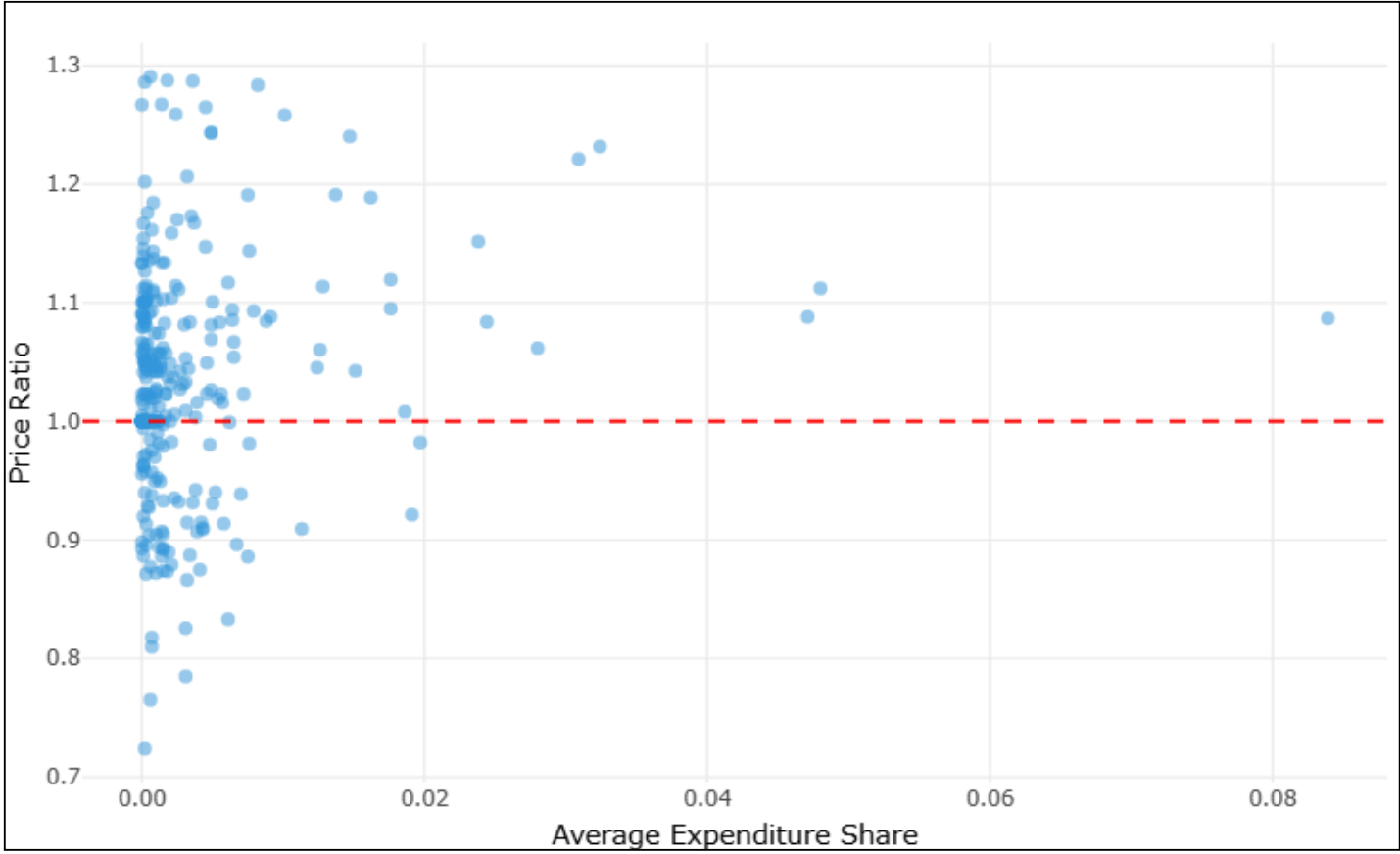
Output data supports main input diagnostics

- Step 4: Price Index calculation and output of contributions

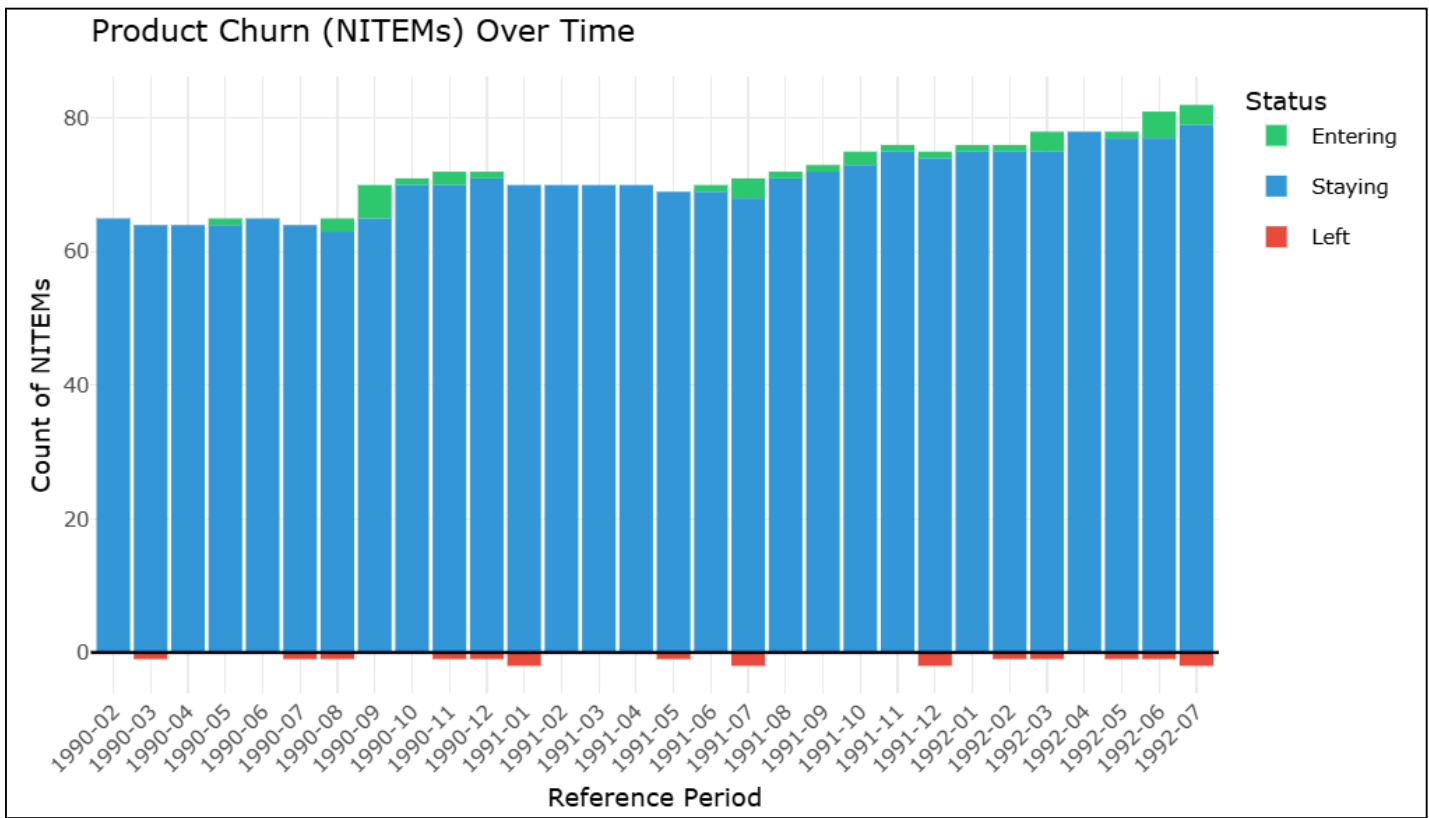
Output data supports both analytics and output diagnostics

5.2 Input diagnostics

Example change in unit price compared to average share



Example churn (entry, stay, exit)



5.3 Analytical/output diagnostics

6 Next steps

Current empirical example focuses on Dominick's dataset and GEKS calculation.

Next steps:

- Expand to TPD and TDH methods
- Expand to additional open datasets
- Validate that dashboards generalize all use cases
- Standardize code into a package (if of value)